

LICENSED ELECTRICIANS ASSESSMENT (LEA)**Licensed Electricians Theory Examination
SAMPLE EXAMINATION "No 2024 – I've had enough" 2026**

Candidate Surname	
Candidate Given Names	

- **PERSONAL NOTEPADS AND PAPER ARE NOT PERMITTED**
- **ONLY PENS MAY BE USED**
- **DO NOT REMOVE ANY SHEETS FROM THIS ASSESSMENT PAPER**
- **PAPERS WITH NO NAME OR SIGNATURE WILL NOT BE MARKED.**
- **UNITS MUST BE SHOWN TO OBTAIN FULL MARKS**



The following reference books are permitted during the assessment session:

AS/NZS 3000:2018 Wiring Rules (including amendment1 & 2)

AS/NZS3012: 2019 Electrical Installations – Demolition and Construction Sites

AS/NZS 3008.1.1:2019 Electrical Installations – Selection of Cables

AS/NZS 4836:2023 Safe Working on or near Low Voltage Electrical installations and Equipment

Electricity Safety (General) Regulations 2019

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	TOTAL
Marks																			

Candidates need to obtain 75% or more to pass this assessment.

Reading Time: 15 Minutes

Working Time: 2 hours

At the end of this time you will be asked to stop.

Candidate Signature: _____

Assessors Signature: _____

WIRING RULES

In the **FIVE** following Wiring Rules questions, you are required to:

- write the Wiring Rules Clause and/or Table number in the space provided
- the correct Wiring Rules Clause and Subclause must be given e.g. 3.5.2(b)(i)

The correct answer to both parts must be given to obtain full marks.

Q 1. (4)

Name 2 of the special arrangements which must be made before a switchboard can be installed within a storage cupboard?

1).....

 2).....

Wiring Rules Clause number.....

[2 + 2 = 4 Marks]

Q 2.

What is the maximum amount of final subcircuits that can be protected by any one Residual Current Device in a residential installation where there is more than one final subcircuit?

.....

Wiring Rules Clause number.....

[2 + 2 = 4 Marks]

Q 3.

In general, the minimum permissible cross-sectional area of insulated copper conductors used for the wiring of socket-outlets is:

Answer:

Wiring Rules Clause number.....

[2 + 2 = 4 Marks]

Q 4.

Where an electrically operated fence is connected directly or indirectly to electricity supply mains, such connections shall be made only through a:

Answer:

.....

Wiring Rules Clause number.....

[2 + 2 = 4 Marks]

Q 5.

What minimum separation do the AS/NZS3000 Wiring Rules recommend regarding the separation between the electrical earthing electrode and nearby gas services?

Answer:

.....

.....

Wiring Rules Clause number.....

[2 + 2 = 4 Marks]

CONSTRUCTION & DEMOLITION SITES

In this question relating to AS/NZS 3012, you are required to:

- write the Standard's Clause number and/or Table number in the space provided
- the correct Clause and Sub-clause number must be given. e.g. 2.10.2 (f)

The correct answer to both parts must be given to obtain full marks.

Q 6.

Where inspection or testing identifies equipment that fails to comply with the criteria given in this standard the equipment shall be-

Answer

(a)

.....

(b).....

.....

AS/NZS3012 Clause Number: _____

[3 + 3 = 6 Marks]

ELECTRICITY SAFETY (GENERAL) REGULATIONS 2019

This question refers to PART 4 – Electrical Installation Work

Amendments to these regulations have been incorporated in these questions.

In the following Regulation question, you are required to:

- * write your answers on the line/s below each question
- * write the complete Regulation and Sub-Regulation number, if applicable, in the space provided. e.g. 401 (e) (3)

The correct answer to parts and must be given to obtain full marks.

Q 7.

Name 2 acceptable materials to be used for the mechanical cover of Consumers mains

(i).....

.....

(ii).....

.....

Answer:

Regulation No:

[2 + 2 = 4 Marks]

ELECTRIC SHOCK SURVIVAL

Q8.

How could help be sought if no response is received from the patient or casualty?

(i).....

.....

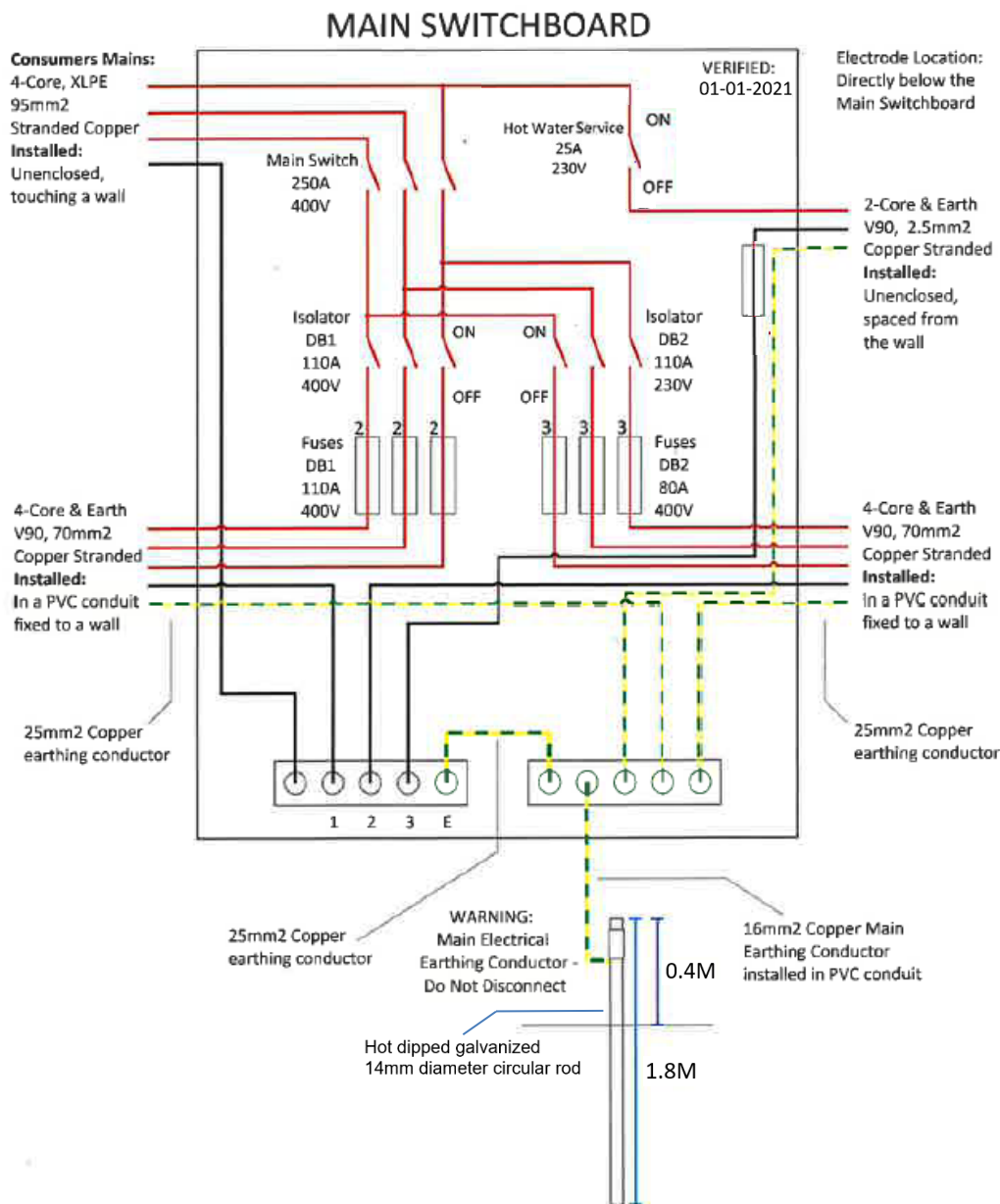
(ii).....

.....

[1 + 1 = 2 Marks]

INSTALLATION DEFECTS - NON DOMESTIC

Q 9.



The MAIN SWITCHBOARD is supplied directly from a pole-mounted transformer via consumers mains installed unenclosed on a cable tray fixed directly to a wall and enters directly into the switchboard enclosure.

It supplies two (2) distribution boards, each having a connected load with a calculated maximum demand of 100A per phase, and a single-phase water heater rated at 25A. The water heater has been deemed not at increased risk of electric shock.

The cables are installed as they are described on the drawing.

Assume the MIMS cables are earthed in accordance with the Wiring Rules and are capable of maintaining supply to the equipment even when exposed to fire and mechanical damage.

All fuses are HRC type.

All screws in bars and links are 80% of the tunnel terminal diameter.

A metal partition separates all switchgear from each other.

The Supply Authority has provided Short Circuit protection of the Point of Supply on the Consumers Mains.

Use the diagram on the previous page.

List **FOUR different defects** together with the contravened Wiring Rules Clause/Table number in the table provided below.

Note: Only the first five defects will be considered.

DEFECT DETAILS	WIRING RULE CLAUSE/TABLE No.

[4 x (2+1) = 12 Marks]

CABLE SELECTION

Q 10.

Three 50 mm² four-core X90 insulated, sheathed and armoured stranded copper cables, including earthing conductors, are connected in parallel to supply a three-phase load.

The cables are protected by a circuit breaker and installed touching in separate conduits in the ground at a depth of 0.5m. The thermal resistivity of the soil is 1.5°C.m/W.

(i) Neglecting voltage drop, what is the **TOTAL** maximum current carrying capacity of the cables which form the circuit?

(ii) What would be the effect on the **TOTAL** current carrying capacity if the cables were spaced 0.45m apart?

	Answer		Answer		
Table 3 (____ ?)		Item			
Table		Column			Answer
Derate/rating table		Column		Factor	
Derate/rating table		Column		Factor	

Answer (i)

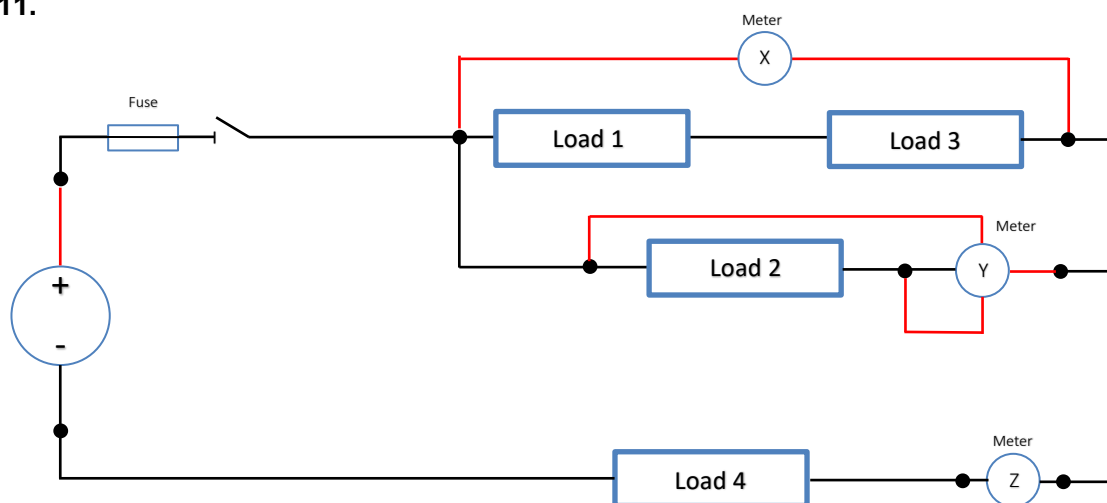
	Answer		Answer		Answer
Derate/rating table		Column		Factor	

Answer (ii)

[2+2+1+1+1+1 = 8 Marks

DC CIRCUITS

Q 11.



For the circuit shown above, you have taken some measurements and recorded the following:

Supply Voltage – 240V

Voltage across load 2 – 160V

Current through load 3 – 4A

Resistance of load 4 – 10Ω

Using these measurements calculate the meter readings for the following meter when the switch is closed:

- (i) meter Z
- (ii) meter Y

Calculate the continuity reading for the following meter when the switch is open:

- (iii) meter X

All calculations including the final answer must be completed to a maximum of **two decimal places**.

Part (i)
Answer:

Part (ii)
Answer:

Part (iii)
Answer:

[3+3+3 = 9 Marks]

MAXIMUM DEMAND**Q 12.**

Calculate the Maximum Demand of a 230V sub-main supplying a distribution board in a non-airconditioned Boarding House installation.

The load connected to the switchboard is:-

- 4 circuits of 4 - 10A double socket outlets
- 1 circuit with 3 - 15A socket outlets
- 2 circuits of 19 -15 watt LED downlights
- 3.0 kw Instantaneous Hot Water system
- 4.5 KW Spa pump heater

All relevant Table details, calculations and units must be shown to obtain full marks.

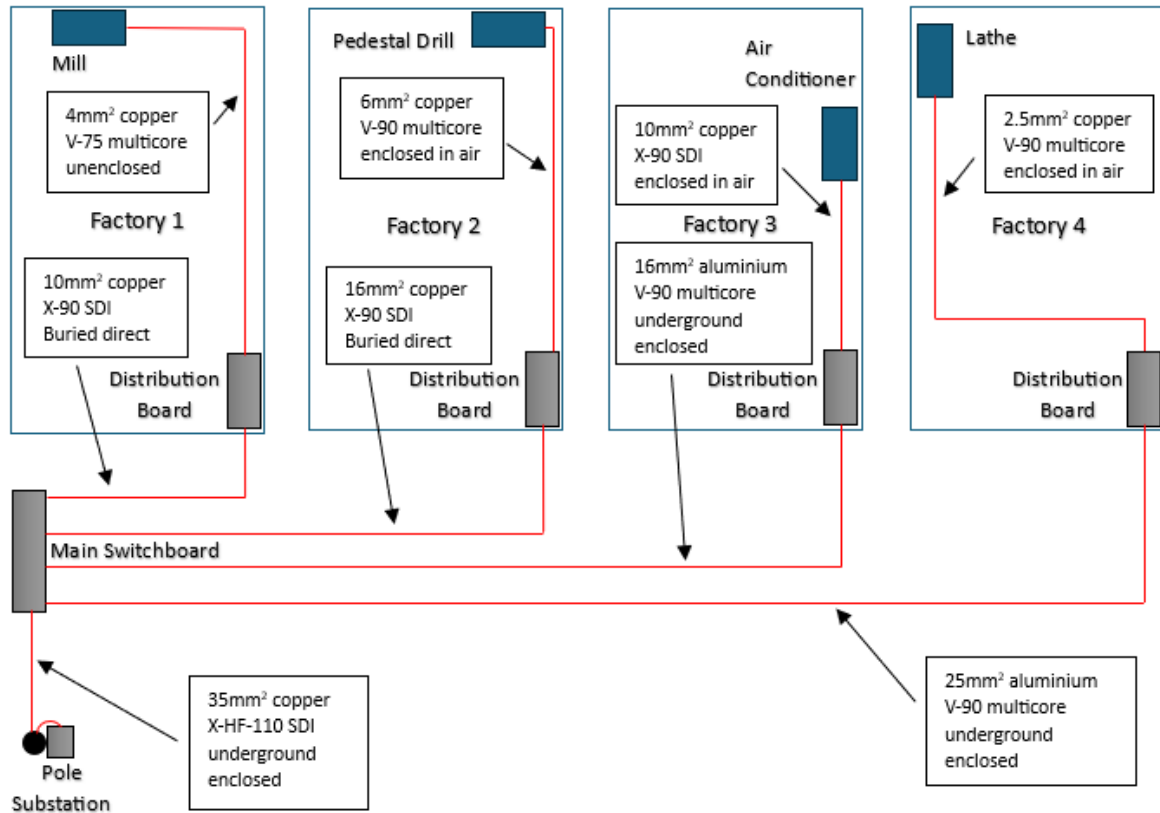
Table		Column	
Equipment	Load Group	Calculation	Maximum Demand
Total Maximum Demand:			

[1 +2+2+2+1 = 8 Marks]

VOLTAGE DROP

Q 13.

In a 400 /230 V, three-phase industrial installation, a three-phase 28A lathe is supplied from a sub-circuit originating at a distribution board in factory 4. The lathe is installed 35 meters away from the distribution board.



Lucky Street

Client: Bob Down	Location	Cable	Distance	Maximum Demand	Voltage
Drawing title: Site plan	Main Switchboard	Consumers Mains	11m	130A	400/230V
Date: 12 February 2025	Factory 1 Distribution board	Sub Mains	14m	57A	400/230V
Location: 65 Lucky St. Brockway	Factory 2 Distribution board	Sub Mains	22m	48A	400/230V
	Factory 3 Distribution board	Sub Mains	34m	60A	400/230V
	Factory 4 Distribution board	Sub Mains	50m	55A	400/230V

Calculate the total voltage drop from the point of supply to the lathe terminals.

All calculations including the final answer must be completed to a maximum of **two decimal places**.

All relevant table details, calculations and units must be show to obtain full marks

Cable	Table	Column	Vc	Calculation	Vd
Consumer Mains					
Sub-mains					
Final Sub-circuit					

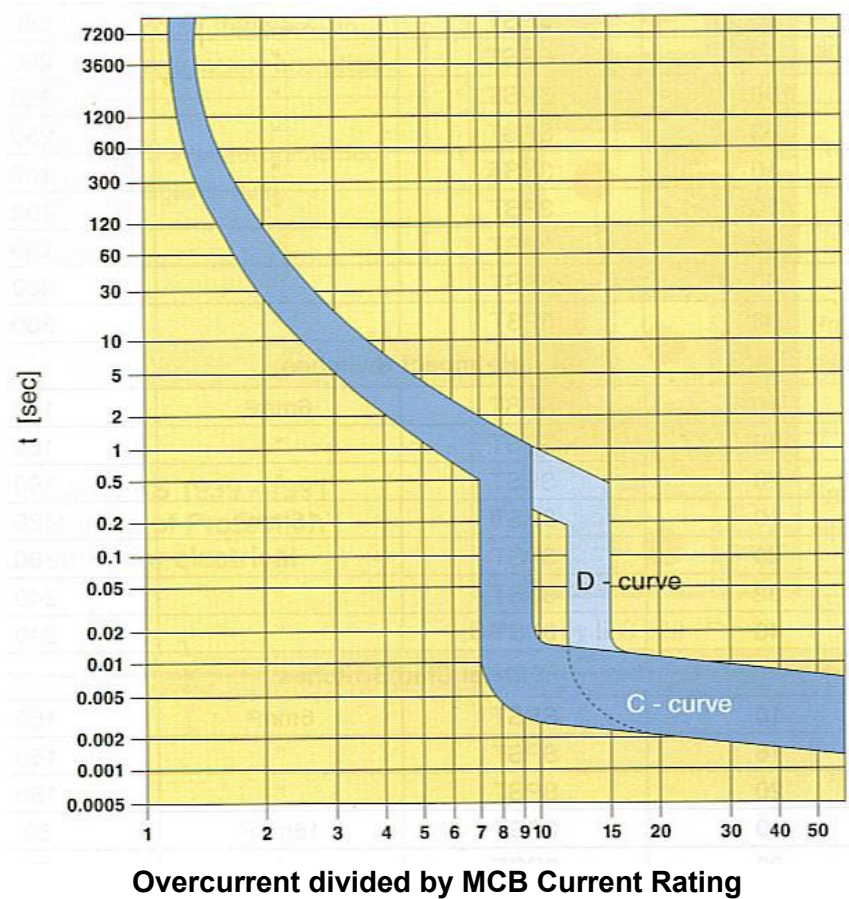
Answer Total Vd:

[3+3+3+1 = 10 marks]

OVERLOAD & SHORT CIRCUIT CALCULATIONS

Q 14.

What are the minimum and maximum tripping times for a 25A Type C miniature overcurrent circuit breaker which is subjected to an over-current of 175A?



Overcurrent divided by MCB current rating:	
Minimum time:	Maximum time:

[1+1+1 = 3 Marks]

OVERLOAD & SHORT CIRCUIT CALCULATIONS
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Q 15.

The main switchboard of a 400/230V industrial installation is directly supplied from a 500KVA transformer which has a prospective fault current of 16725 A per phase.

Submains supply a distribution board from the main switchboard.

The following information is known:-

- Impedance of the Consumers Mains = 0.00310Ω
- Impedance of the Submains cables = 0.02314Ω

Determine the prospective fault current at:-

- the main switchboard; and
- the distribution board.

Work impedances to 5 decimal places.

All calculations must be shown to obtain full marks.

Transformer Impedance:	
MainSw/Bd:	
Dist/Bd:	

[3+3+3 = 9 Marks]

RESIDUAL CURRENT DEVICES

Q 16.

Is it permissible for the earthing conductor or protective bonding conductor to pass through the magnetic circuit of an RCD?

Answer.....

.....

Wiring Rules Clause number.....

[3 Marks]

MOTORS AND STARTERS

Q 17.

CIRCLE the letter in front of the statement you consider to be the most correct.

Which of the following starters will give a three phase motor the best starting torque

- A Primary Resistance starter
- B Direct-on-line
- C Secondary Resistance Starter
- D Star Delta starter

[2 Marks]

AS/NZS 4836:2011

Q 18.

When performing isolation, what information should each personal padlock identify?

.....

.....

.....

AS/NZS 4836. Clause number.....

[2 + 2 = 4 Marks]